

LING4136

Buddy Challenge

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Scientific claim



**"Moving your hands
in meaningful ways
can benefit cognitive load"**

Why?

A		E	
	B		
	D		C









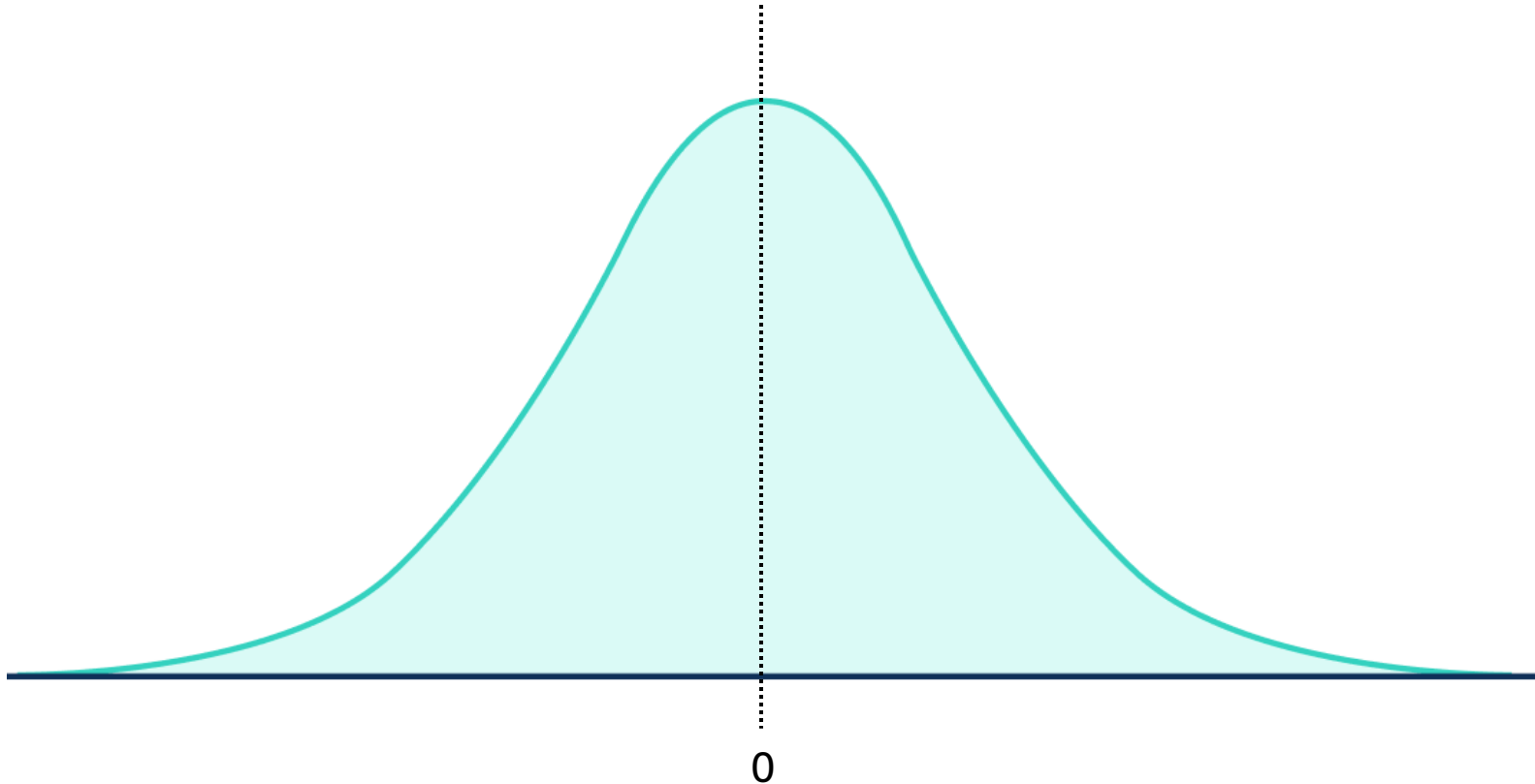
Reproducibility

Computational reproducibility
was good :D

Analytical reproducibility
was mid :/

Bayesian Analysis

We used weakly informative priors centred on 0.



SESOI

5% effect on recall

Langerock (2025)

Rohrer (2020)

Goldin-Meadow (2021)

= 0.25 of a letter

Bayes Factor Interpretation Scheme

Lee & Wagenmakers (2014)

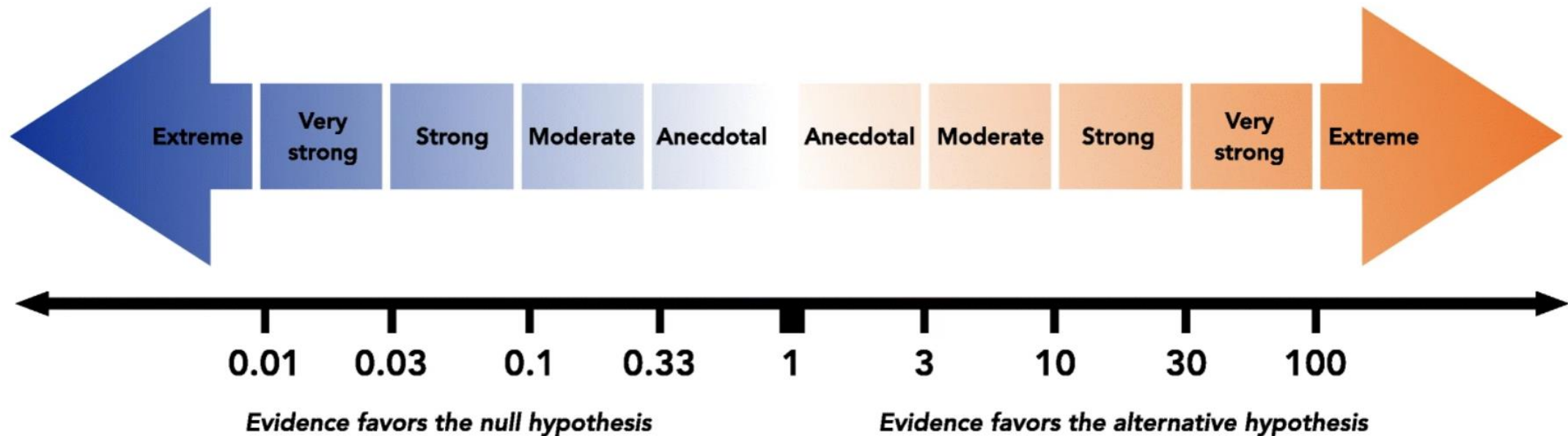
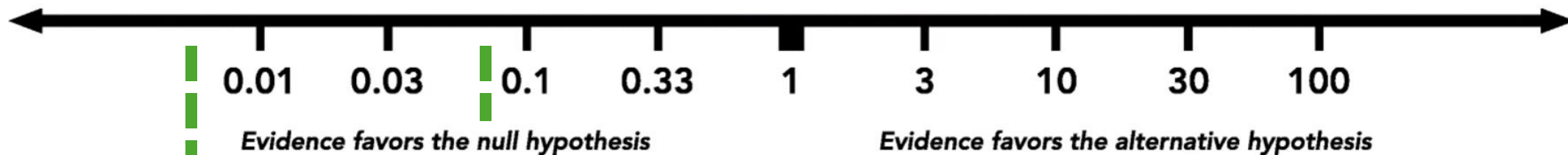
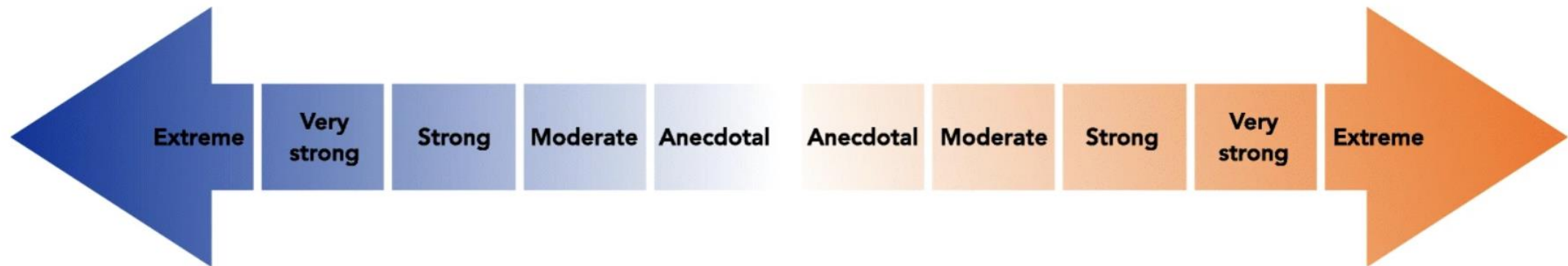


Image: Timo Roettger (2026)



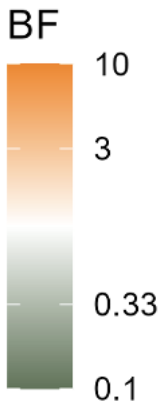
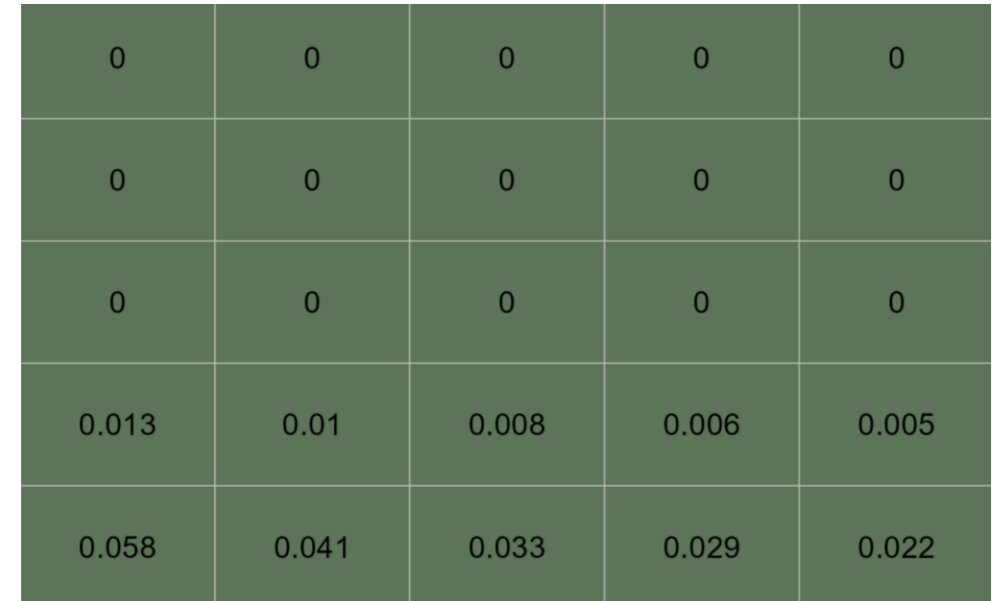
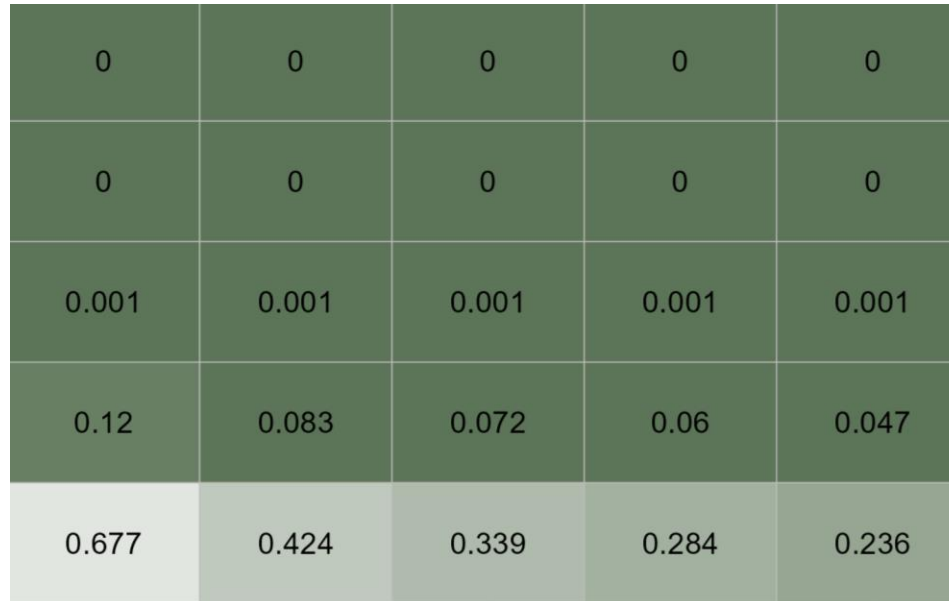
letters ~ hands still

all other predictors

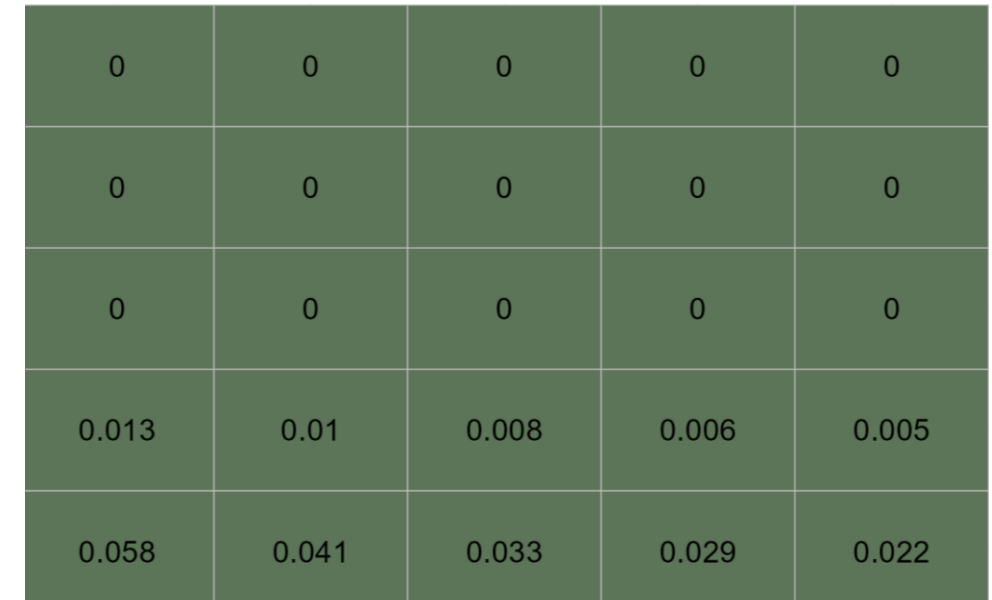
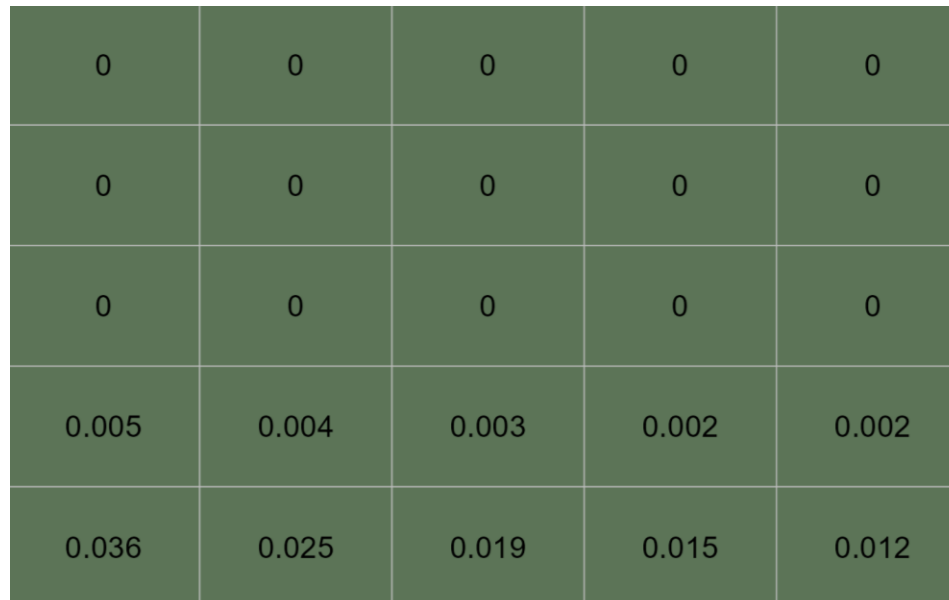
Hands Still Condition

Make Condition

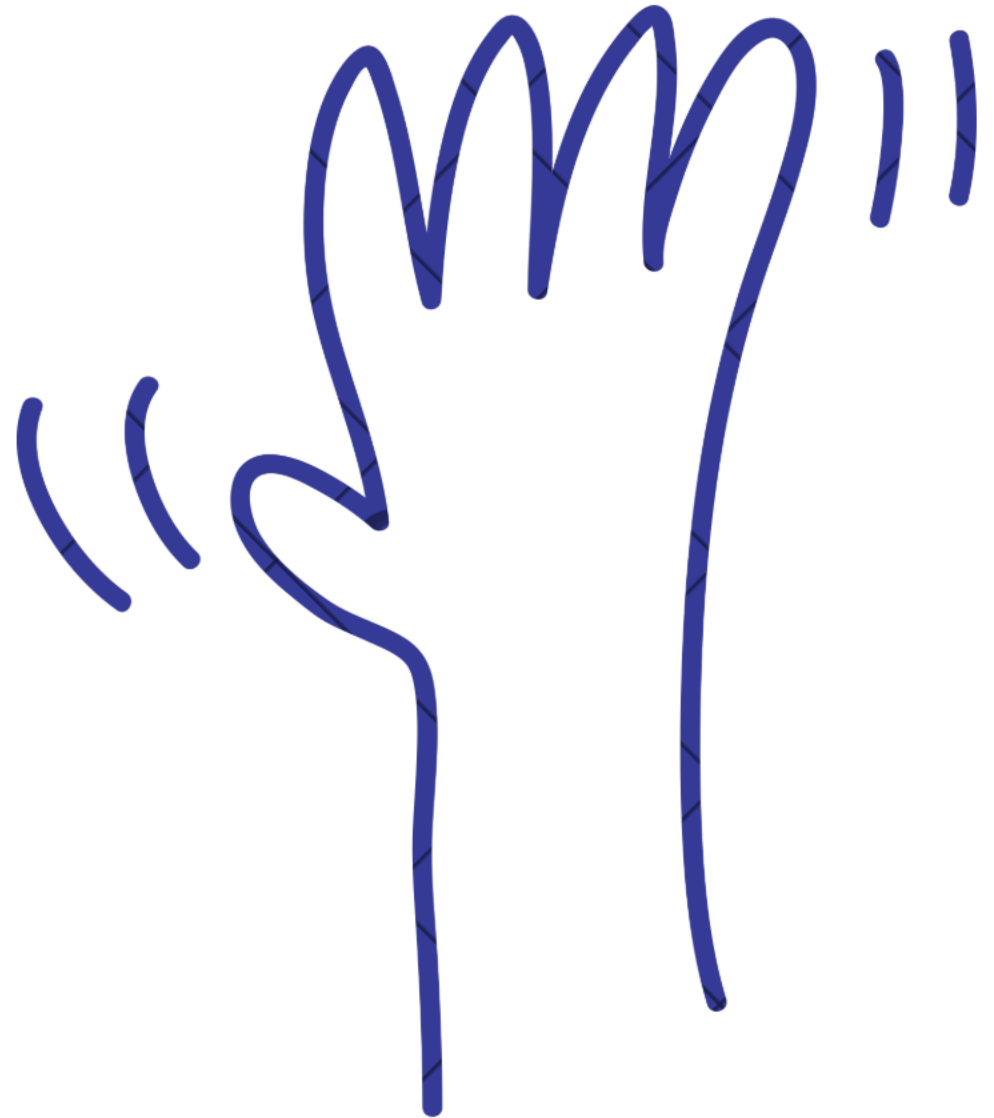
Letter Recall



Location Recall



Thanks!



References

- Goldin-Meadow, S., Nusbaum, H., Kelly, S. D., & Wagner, S. (2001). Explaining Math: Gesturing Lightens the Load. *Psychological Science*, 12(6), 516–522. <https://doi.org/10.1111/1467-9280.00395>
- Langerock, N., Oberauer, K., Throm, E., & Vergauwe, E. (2025). The cognitive load effect in working memory: Refreshing the empirical landscape, removing outdated explanations. *Journal of Memory and Language*, 140, 104558. <https://doi.org/10.1016/j.jml.2024.104558>
- Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian Cognitive Modeling: A Practical Course*. Cambridge University Press.
- Rohrer, P. L., Delais-Roussarie, E., & Prieto, P. (2020). Beat Gestures for Comprehension and Recall: Differential Effects of Language Learners and Native Listeners. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.575929>